

HH Goa 2026 Shortlisting Task 2: Build a Voice-Enabled RAG Model

What to build

A **voice-enabled Retrieval-Augmented Generation (RAG) system** — a user speaks a question, your pipeline transcribes it, retrieves relevant context from a provided dataset, and returns an answer, end to end.

Pipeline shape: **Voice input** → **Speech-to-text** → **Chunking/Retrieval (vector DB)** → **Answer generation**

Dataset

We will provide the dataset to build your RAG pipeline on:
<https://huggingface.co/datasets/ai4bharat/MSMARCO-XI>

Technical requirements

1. Speech-to-text

Use **either Sarvam or ElevenLabs** for voice-to-text. Pick one.

2. Chunking

Chunking strategy should be **vast** — don't submit a single naive fixed-size chunking approach. We want to see real thought put into how the dataset is split, indexed, and retrieved (e.g. multiple chunking strategies, overlap handling, semantic vs. fixed-size splitting, metadata-aware chunking, etc.).

3. Latency target

The full process — chunking + vector DB retrieval + everything through to final output — should complete in **under 50ms**.

4. Latency analytics

Submit **P50 / P70 / P100 latency numbers** for your pipeline, measured across a reasonable number of test queries — not a single best-case run.

5. Harness your model

Your model/pipeline should be run inside a proper harness — structured orchestration around the model (tool calls, retries, structured input/output handling, error recovery) rather than a single raw prompt-in, text-out call.

6. Guardrail your model

Add guardrails around your model — handling for off-topic queries, unsafe/inappropriate inputs, hallucination checks, or answers not grounded in the retrieved context. Show that your system knows when *not* to answer, not just how to answer.

Submission requirements

- Fill the submission form: <https://forms.gle/MNvCjcv23Hn2Eeu58>
- GitHub repo link
- Live working link
- 2 videos (see below)

No resubmissions will be allowed — submit only when your build is final.

Video 1 — Team/process video

- **90 seconds**
- Shows how your team is working on this — process, not the product itself.

Video 2 — Demo video

- Demo of the actual project working end to end.
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Promotion requirement (mandatory)

Both videos must be uploaded to **Instagram, X, and LinkedIn** — **by every individual team member**, not just one shared team post. At least 1 **Instagram** account should be public.

Every post, on every platform, by every member, must include:

#RAGInGoa

Timeline

- **Task launch:** August 13, 2026
 - **Deadline:** August 22, 2026, 11:59 PM
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